$t \qquad = \frac{1}{2}\sin 2t = \frac{1}{2}\cdot\frac{2}{s^2+4} = \frac{1}{s^2+4}$

1.(1) $L(f(t)) = L(\sin t)\,\cancel{L(\cos t) = \frac{1}{1+s^2}}\cdot\frac{s}{1+s^2} = \frac{s}{(1+s^2)^2}$

$\cancel{3(3)\ F(s) = \frac{1}{s^2} - \frac{1}{s^3} + \frac{1}{s^4} - \frac{1}{s^5}}$

$\cancel{L^{-1}F(s) = L^{-1}\left(\frac{1}{s^2}\right) = \mathrm{Res}\left[\frac{(s^3-s^2+s-1)e^{st}}{s^5}, 0\right]^{4阶} = \frac{1}{4!}\left(s^3-s^2+s-1\right)e^{st}\Big|^{(4)}_{s=0}}$

$\cancel{= \frac{5}{24}e^{t} = \frac{5}{8}e^{t}}$

3(3) $\cancel{L^{-1}[F(s)] = \mathrm{Res}[F(s)e^{st}, 0]^{4阶} = \frac{1}{4!}[F(s)e^{st}]^{(4)}\Big|_{s=0}}$

$\cancel{= \frac{1}{4!}[(s^3-s^2+s-1)e^{st}]^{(4)}\Big|_{s=0} =}$

$L^{-1}[F(s)] = L^{-1}\left[\frac{1}{s^2}\right] - L^{-1}\left[\frac{1}{s^3}\right] + L^{-1}\left[\frac{1}{s^4}\right] - L^{-1}\left[\frac{1}{s^5}\right] = t - \frac{t^2}{2} + \frac{s^4}{6} - \frac{s^5}{24}$

5. (5) $\cancel{L[f(t)] = u(L[t^3] - L[t^2])} \qquad L(t^3) = \frac{\Gamma(4)}{s^4} = \frac{6}{s^4} \qquad L(t^2) = \frac{\Gamma(3)}{s^3}$

$\cancel{= u\left(\frac{6}{s^4} - \frac{2}{s^3}\right)} \qquad u(t-1) \to \frac{1}{s}e^{-s} \qquad t^2u(t-1) \to \left(\frac{1}{s}e^{-s}\right)'' = \left(\frac{2}{s^3} + \frac{1}{s^2} + \frac{1}{s}\right)e^{-s}$

(7) $L[1] = \frac{1}{s}$

$L[t^n] = (-1)^n\left(\frac{1}{s}\right)^{(n)} = n!\,\frac{1}{s^{n+1}}$

$L[(t+1)^n] = \cancel{e^{-s}\,n!\,\frac{1}{s^{n+1}}}$ 时移只能向右平移!!! 二项式展开

7.(2) $\int_0^\infty \frac{f(t)}{t}dt = \int_0^\infty F(s)ds$

$F(s) = L[e^{-at} - e^{-bt}] = \frac{1}{s+a} - \frac{1}{s+b}$

$\therefore LHS = \int_0^\infty \frac{1}{s+a} - \frac{1}{s+b} = \ln\frac{b}{a}$

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8(2)

$$F(s)=L[f(t)]=-\left[L\left(\int_0^t e^{-x}\sin x\,dx\right)\right]'$$

$$L\left[\int_0^t e^{-x}\sin x\,dx\right]=\frac{1}{s}L[e^{-x}\sin x]=\frac{1}{s}\cdot\frac{1}{(s+1)^2+1}$$

$$L[e^{-x}\sin x]=F(s+1)=\frac{1}{(s+1)^2+1}$$

$$F(x)=L(\sin x)=\frac{1}{s^2+1}$$

$$\therefore F(s)=-\left[\left(\frac{1}{s}\frac{2(s+1)}{[(s+1)^2+1]^2}\right)+\left(-\frac{1}{s^2}\frac{1}{(s+1)^2+1}\right)\right]=\frac{2(s+1)}{s[(s+1)^2+1]^2}+\frac{1}{s^2}\cdot\frac{1}{(s+1)^2+1}$$

$e^{-s}\,e^{st}$

$$9(4).\ L^{-1}(F(s))=\sum_{k=1}^{n}\mathrm{Res}\left[\frac{e^{s(t-1)}}{s(s^2+1)},s_k\right]$$

$$\mathrm{Res}\left[\frac{e^{s(t+1)}}{s(s^2+1)},0\right]=1$$

$$\mathrm{Res}\left[\frac{e^{s(t+1)}}{s(s^2+1)},i\right]=\frac{e^{i(t+1)}}{-2}=-\frac{i}{2}\sin(t+1)$$

$$=-\frac{1}{2}[\cos(t+1)+i\sin(t+1)]$$

$$\mathrm{Res}\left[\frac{e^{s(t+1)}}{s(s^2+1)},-i\right]=\frac{e^{-i(t+1)}}{-2}=\frac{i}{2}\sin(t+1)$$

$$=-\frac{1}{2}[\cos(t+1)+i\sin(t+1)]$$

$$1-\cos(t-1)$$

$$L^{-1}\left(\frac{e^{-s}}{s}\right)-L^{-1}\left(e^{-s}\frac{s}{s^2+1}\right)$$

$$=u(t-1)-\cos(t-1)\,u(t-1)$$

$$11(5).\ L[y'']=s^2L(y)-sy(0^+)-y'(0^+)=s^2L(y)-3s+2$$

$$\therefore (s^2+4)L[y]-3s+2=\frac{1}{s},\ 0\le t\le 4\quad \int_0^4 e^{-st}\,dt=\frac{1}{s}(1-e^{-4s})$$

$$\Rightarrow L[y]=\frac{(1-e^{-4s})+3s^2-2s}{s(s^2+4)}=\frac{-e^{-4s}}{s(s^2+4)}+\frac{1}{s(s^2+4)}+\frac{3s-2}{(s^2+4)}$$

$$\therefore y=\sum_{k=1}^{n}\mathrm{Res}[L[y],s_k]$$

$$=\frac{1}{4}(1-\cos 2t)+3\cos 2t-\sin 2t$$

$$-\frac{1}{4}[1-\cos 2(t-4)]u(t-4)$$

$$\mathrm{Res}[L[y],2i]=\frac{1-e^{-8i}-12-4i}{-8}=\frac{11+4i+e^{-8i}}{8}$$

$$\mathrm{Res}[L[y],-2i]=\frac{1-e^{8i}-12+4i}{-8}=\frac{11-4i+e^{8i}}{8}$$

$$\mathrm{Res}[L[y],0]=\frac{0}{4}$$

$$\therefore y=\frac{22+e}{8}$$

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